

Saksham Sharma

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Education

Thapar Institute of Engineering and Technology
Bachelor of Technology in Electronics and Computer Engineering

Patiala, Punjab
Aug 2023 – May 2027

Projects

RAG + LangChain Pipeline (Async, Queue-Based) 2026
Python, LangChain, Qdrant, FastAPI, RQ, Redis/Valkey, Docker, Gemini API

- Built a RAG pipeline with Gemini-grounded retrieval and generation using Qdrant as the vector database.
- Implemented batched ingestion and tuned chunk overlap to reliably index the full corpus at scale within API rate limits.
- Re-architected the synchronous pipeline into a production-style async job queue: FastAPI enqueues via `/chat`, an RQ worker (Redis/Valkey) processes jobs, `/job-status` polls results – orchestrated with Docker Compose for horizontal worker scaling.

URL Shortener API with Click Analytics July 2026
Python, FastAPI, PostgreSQL (Neon), SQLAlchemy, pytest, Alembic, async/await

- Engineered a production-grade REST API with collision-free short-code generation, real-time click analytics, and async SQLAlchemy; deployed on Railway with PostgreSQL (Neon).
- Built a 13-test suite with pytest + pytest-asyncio covering core functionality, connection pooling, and edge cases.
- Designed a normalized PostgreSQL schema with indexed columns, foreign-key constraints, and Alembic migrations; prevented SQL injection via parameterized queries and removed manual `create_all()` from the app lifespan.

Mini ATS – Candidate Pipeline Tool July 2026
React, TypeScript, Tailwind CSS, ASP.NET Core, Entity Framework Core, SQLite, Docker

- Built and deployed a full-stack applicant tracking system with job posting, PDF/DOCX resume upload, and a 5-stage recruiter pipeline (Applied to Rejected/Offer); frontend on Vercel, backend on Railway via Docker.
- Designed a 5-endpoint REST API in ASP.NET Core (Minimal API) with EF Core, a normalized Job–Candidate relational model, cascading status updates, and multipart file-upload handling.
- Made deliberate architectural trade-offs (SQLite for iteration speed, Minimal API over MVC) to prioritize a fully working, well-documented system over unfinished complexity.

Technical Skills

Languages: Python, C/C++, JavaScript, TypeScript, SQL, HTML5, CSS3

Backend: FastAPI, Flask, ASP.NET Core, Node.js, Express, REST APIs, async/await, concurrency

AI/ML: LangChain, RAG pipelines, Qdrant, Gemini API, embedding models, prompt-driven retrieval

Databases: PostgreSQL, SQLite, SQLAlchemy, Entity Framework Core

Infra/Tools: Docker, Docker Compose, Redis/Valkey, RQ, Git, Railway, Vercel, pytest, Alembic

Core CS: Data Structures, Algorithms, OOP, System Design, Concurrency, Security

Leadership and Extracurricular

Vice President, TED-Ed Club – Spring Dale Senior School (1-year term)

Participant, Smart India Hackathon (SIH)